

Dhia Problem 3 — PROMPT 6 results ($\sigma_1 = \sigma_2 = 0.1$)

Skin-friction coefficient $f''(0)$ and heat-transfer rate $-t'(0)$ versus the stretching/shrinking parameter λ for three inclination angles $\alpha = 0, \pi/4, \pi/2$, with the parameters $\sigma_1 = \sigma_2 = 0.1$, $\phi_1 = \phi_2 = 0.1$, $\beta_v = \beta_T = 0.5$, $s = 1$, $Pr = 6.2$, $Ec = 1$, $m = 1$, $\varepsilon = 1$, $L = 1$. Each curve figure carries an inset that zooms the dual-solution region near the fold, and an arrow marking the direction of increasing α . The second (lower) solution is continued up to $\lambda = 0$ where it exists. Dual velocity and temperature profiles are shown at $\lambda = -0.38$ ($Gr = 0.1$) and $\lambda = -0.30$ ($Gr = -0.5$), each inside the dual-solution window of all three angles, with the fluid (nanofluid) and dust phases in separate figures. Both the buoyancy-assisting case $Gr = 0.1$ and the opposing case $Gr = -0.5$ are reported.

Gr	quantity	$\alpha = 0$	$\alpha = \pi/4$	$\alpha = \pi/2$
0.1	fold λ_c	-0.44052	-0.43426	-0.42000
0.1	2nd sol. reaches λ	-0.2297	0.0000	0.0000
-0.5	fold λ_c	-0.33931	-0.36041	-0.42000
-0.5	2nd sol. reaches λ	0.0000	0.0000	0.0000

Table 1. Critical (fold) values and the smallest $|\lambda|$ reached on the second branch. Where the value is 0.0000 the second solution was continued to $\lambda = 0$.

Gr = 0.1 (buoyancy-assisting)

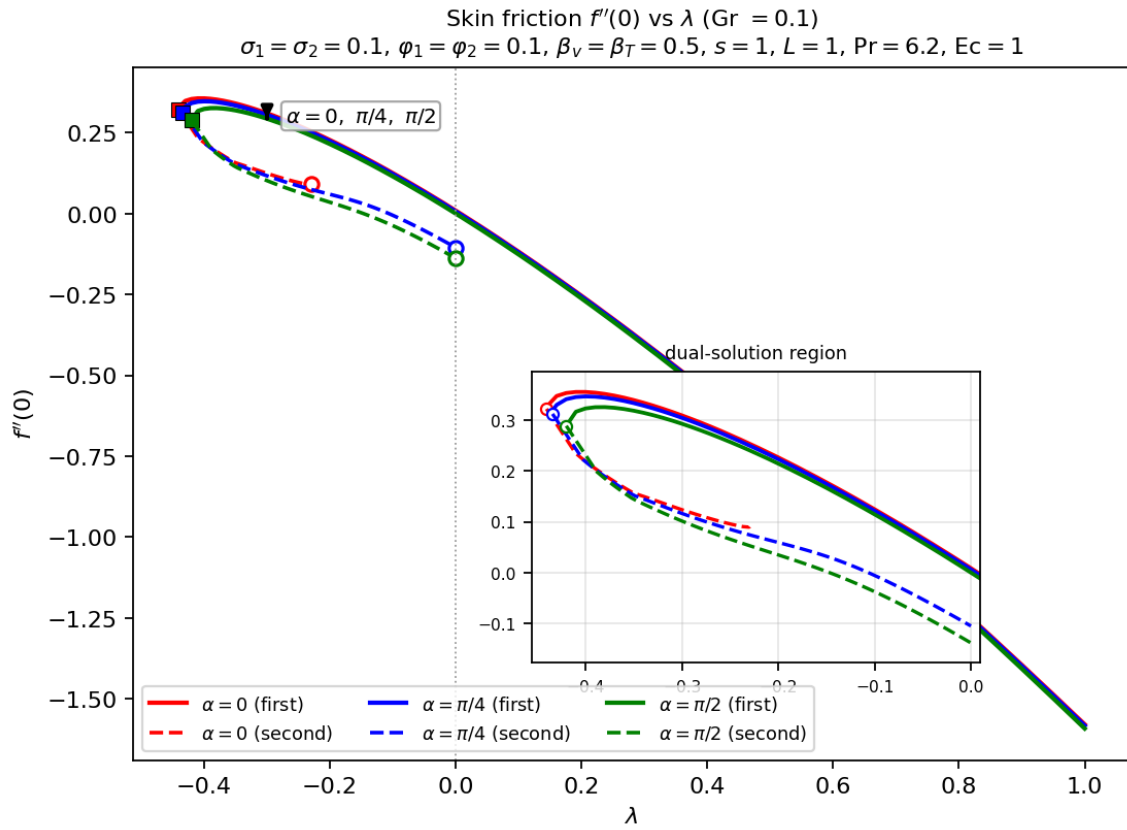


Figure. $f''(0)$ vs λ (first: solid, second: dashed; square = fold, open circle = 2nd-branch end). Inset: dual-solution region; arrow: increasing α .

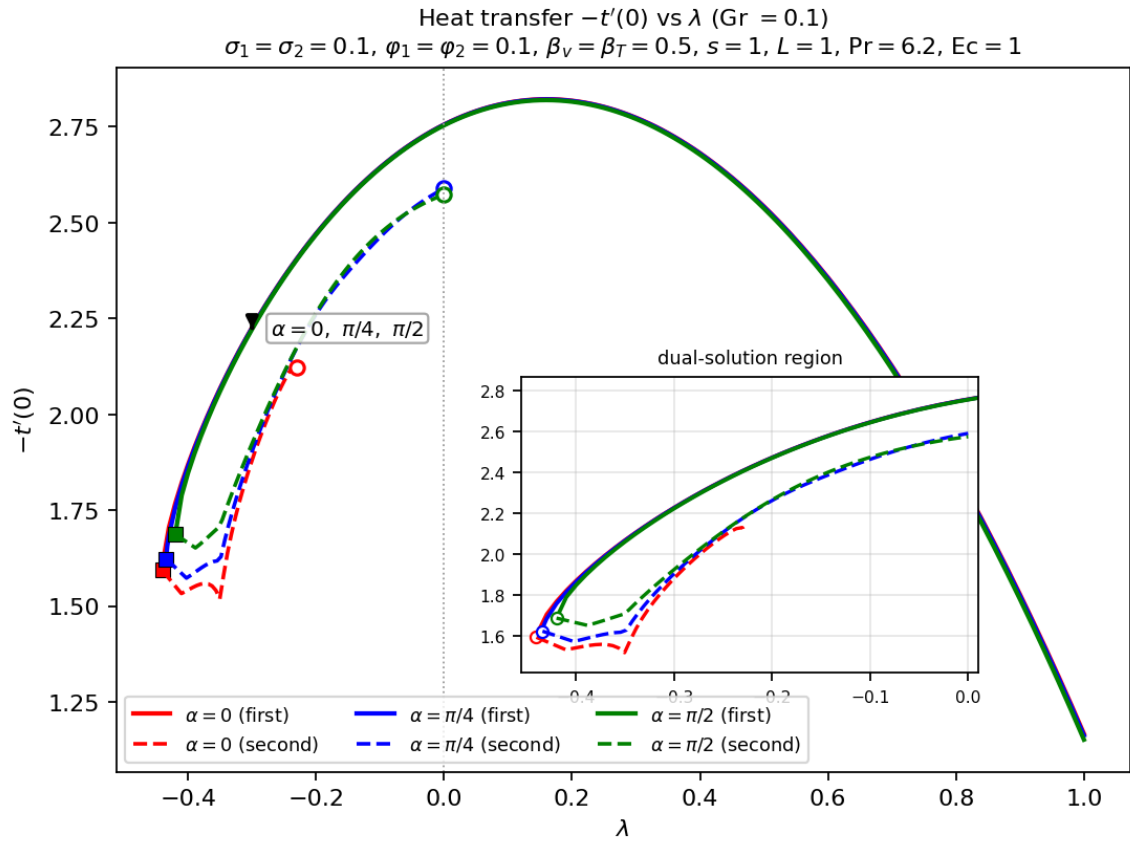


Figure. $-t'(0)$ vs λ (Nusselt number). Inset: dual-solution region; arrow: increasing α .

Dual profiles at $\lambda = -0.38$ — Gr = 0.1 (buoyancy-assisting)

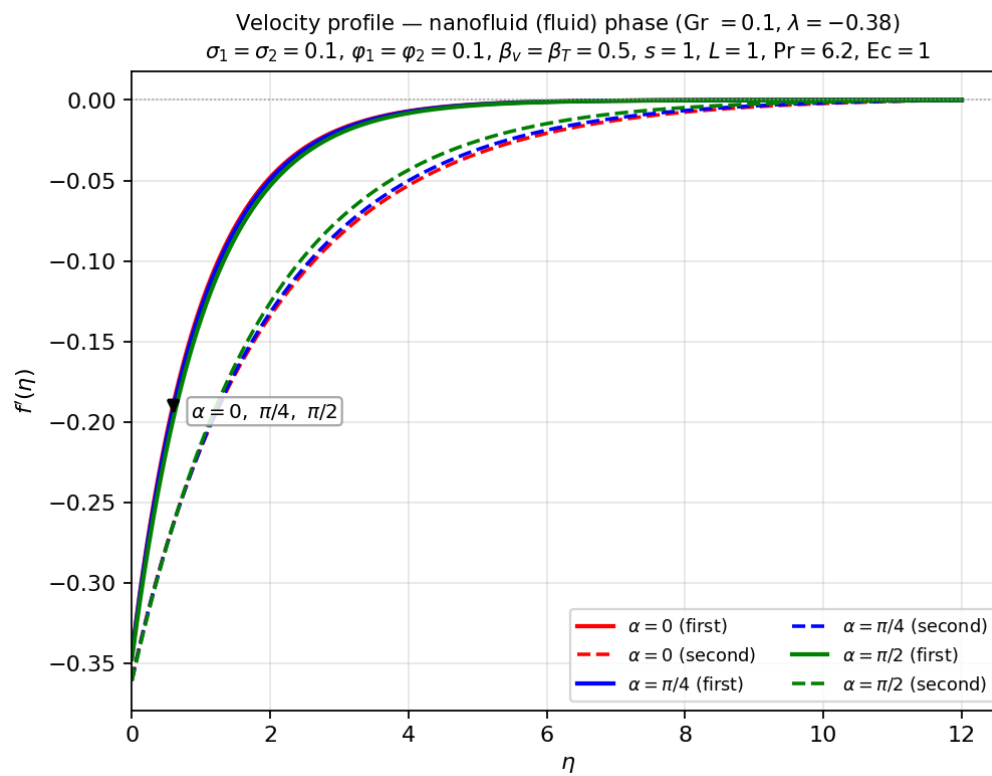


Figure. Fluid-phase velocity $f'(\eta)$.

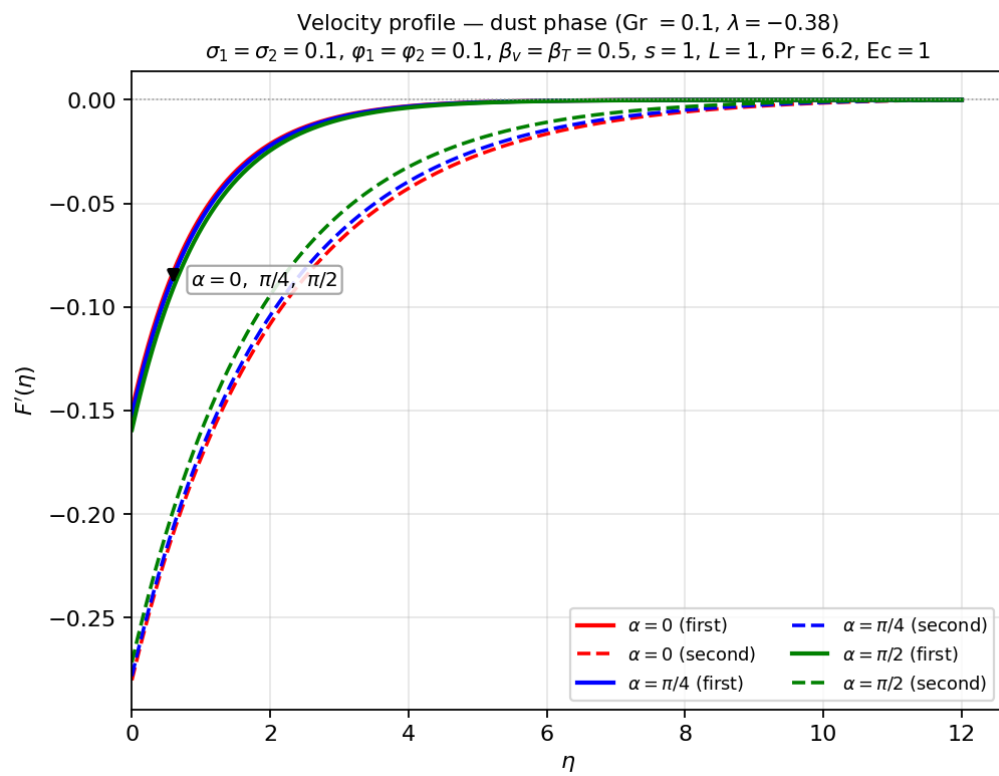


Figure. Dust-phase velocity $F'(\eta)$.

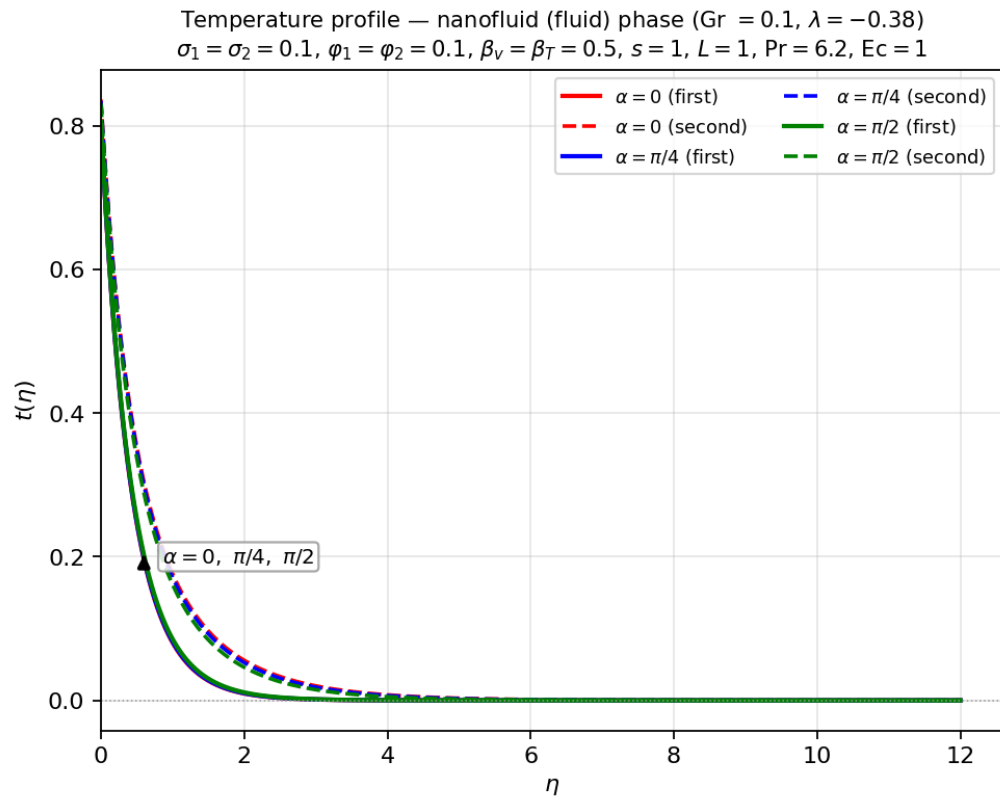


Figure. Fluid-phase temperature $t(\eta)$.

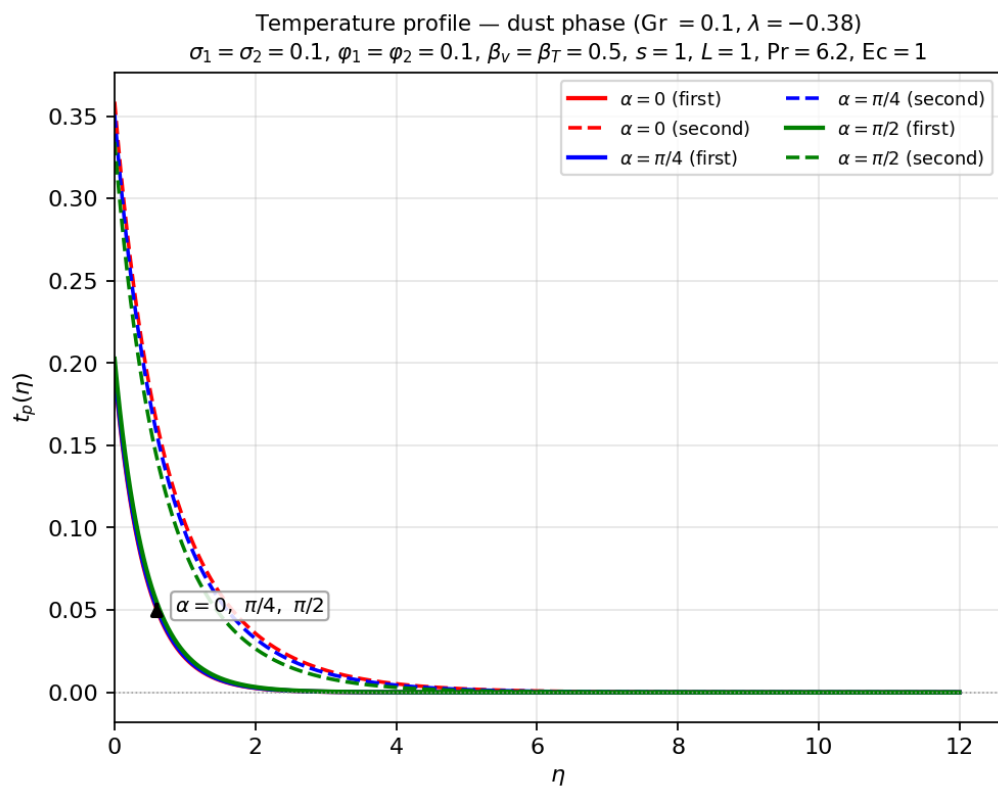


Figure. Dust-phase temperature $t_p(\eta)$.

Gr = -0.5 (opposing)

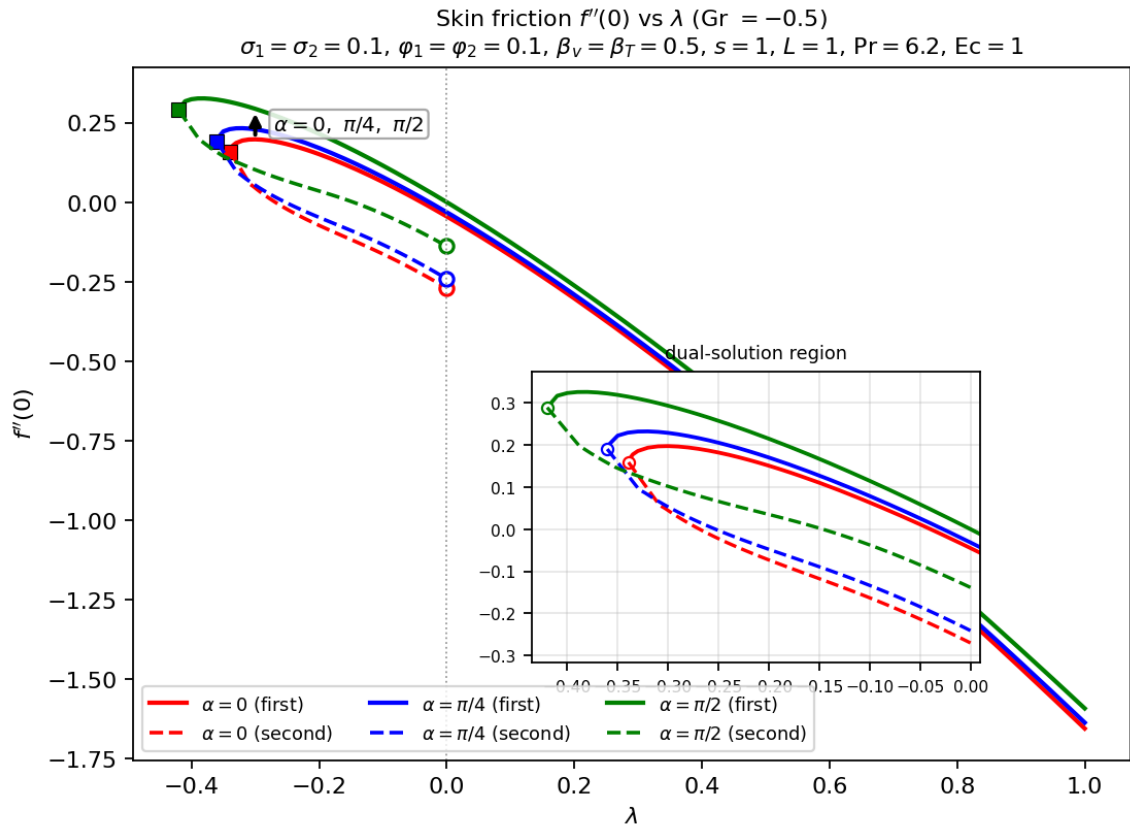


Figure. $f''(0)$ vs λ (first: solid, second: dashed; square = fold, open circle = 2nd-branch end). Inset: dual-solution region; arrow: increasing α .

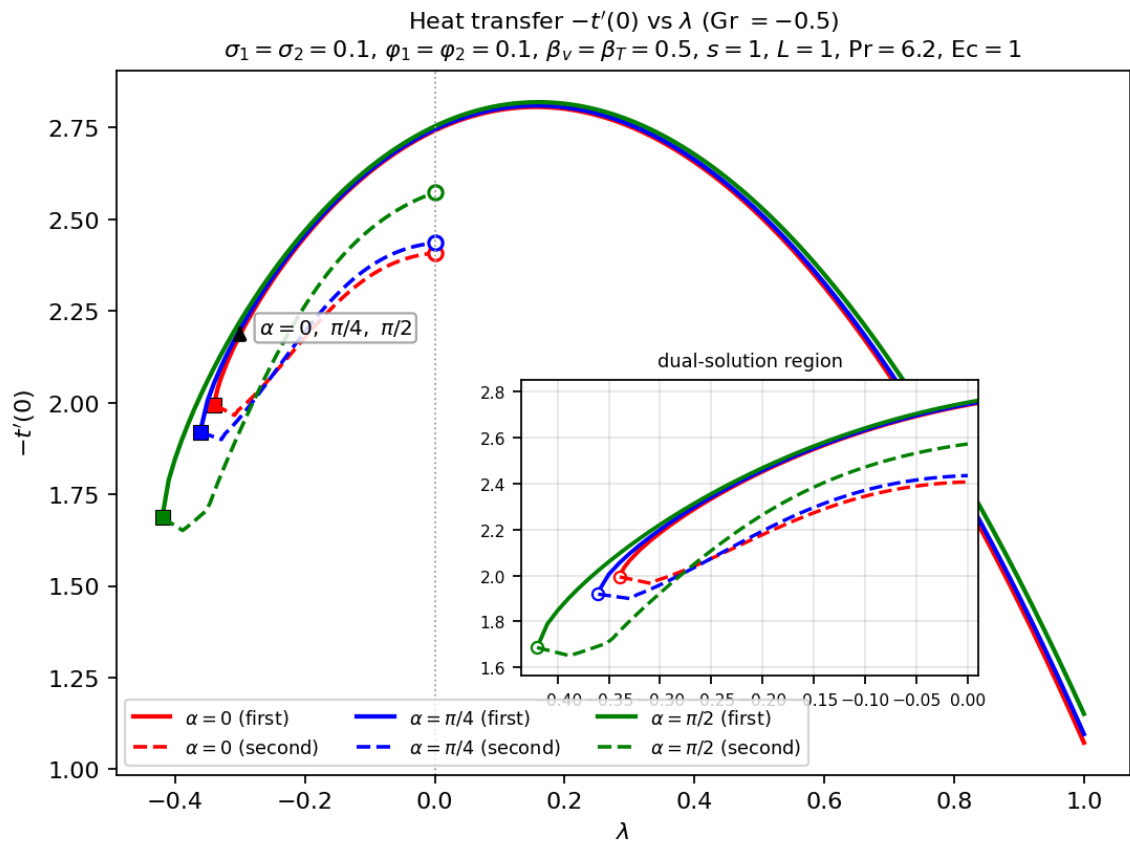


Figure. $-t'(0)$ vs λ (Nusselt number). Inset: dual-solution region; arrow: increasing α .

Dual profiles at $\lambda = -0.3$ — $Gr = -0.5$ (opposing)

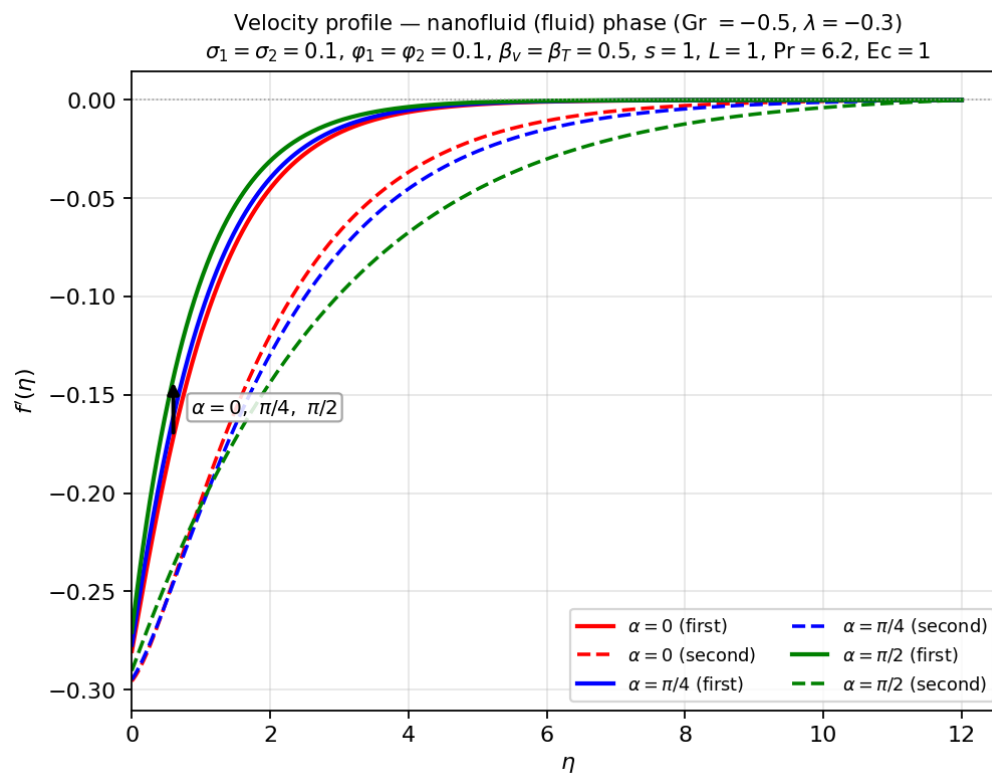


Figure. Fluid-phase velocity $f'(\eta)$.

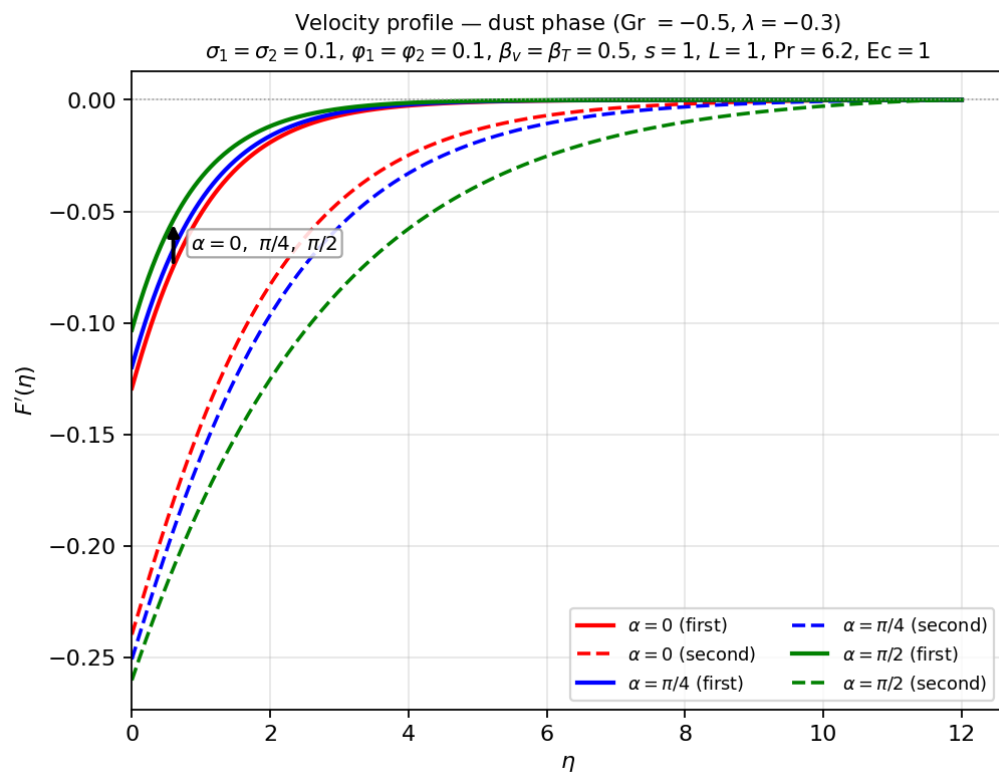


Figure. Dust-phase velocity $F'(\eta)$.

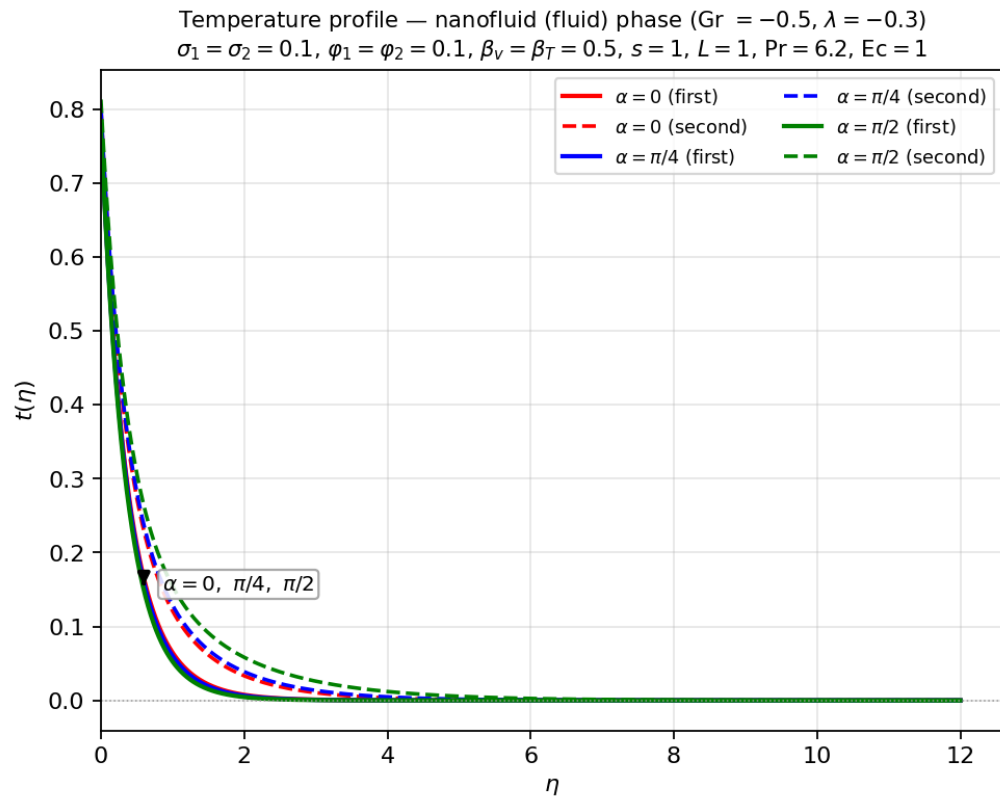


Figure. Fluid-phase temperature $t(\eta)$.

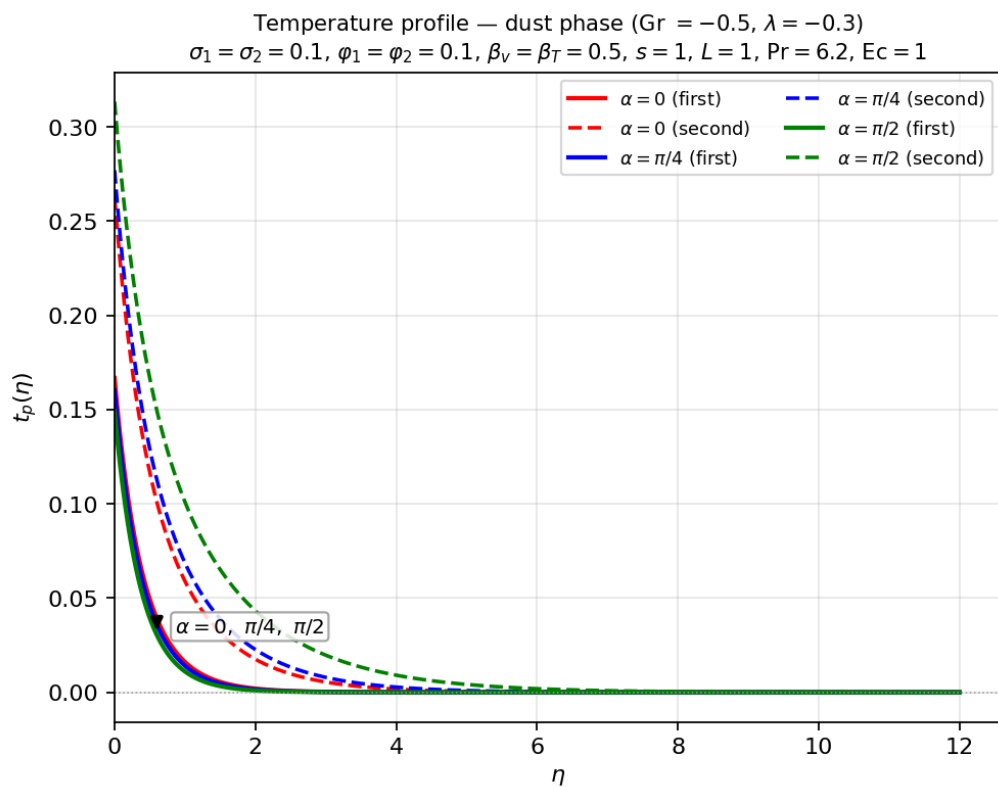


Figure. Dust-phase temperature $t_p(\eta)$.